

# Proposal: Replace the sourcing reverse block scan with an index lookup

|           |                                                                                                                                                        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                                                                                                                                        |
| Status    | Proposal                                                                                                                                               |
| Area      | JetStream stream sourcing ( <code>server/stream.go</code> ) + store index ( <code>server/filestore.go</code> , <code>server/memstore.go</code> )       |
| Date      | 2026-06-06                                                                                                                                             |
| Companion | <code>sourcing-durable-resume.md</code> (lifecycle, durable consumer, triggers)                                                                        |
| Target    | <code>startingSequenceForSources</code> ( <code>stream.go:4694</code> ) and <code>setStartingSequenceForSources</code> ( <code>stream.go:4566</code> ) |

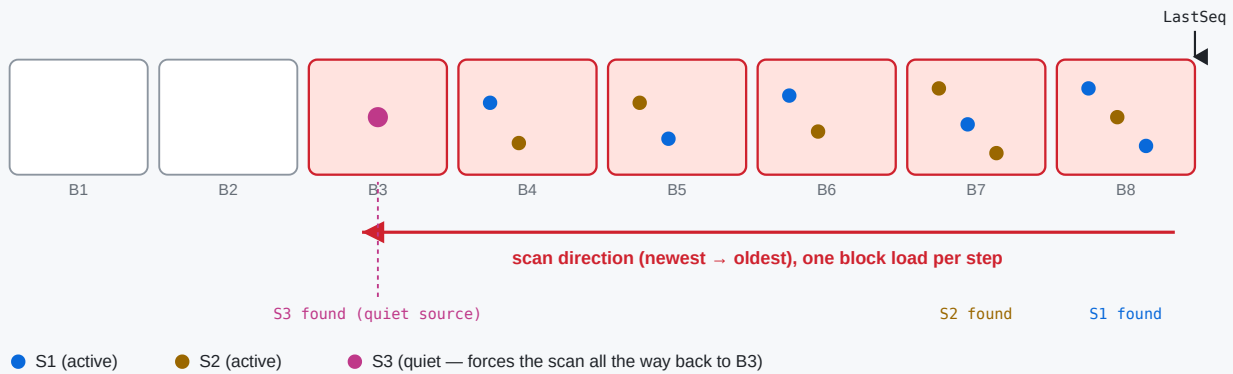
## 1. The problem in one paragraph

On hub-stream **leader election / restart**, `setupSourceConsumers` calls `startingSequenceForSources` **unconditionally** ( `stream.go:4821` ). That function reconstructs each source's last sourced sequence by walking the stream's **own store backwards from `LastSeq`**, one block at a time ( `LoadPrevMsgMulti` , `filestore.go:9425` ), reading the `Nats-Stream-Source` header off matching messages, until **every** source has been located.

Note: the walk is not as naive as message-by-message — within each block the per-block index ( `fss` ) skips non-matching messages, and the per-source sublist narrows as sources are found. But `LoadPrevMsgMulti` still **visits every block** from `LastSeq` back to the match (loading/decompressing cold ones), and the outer loop **re-walks** as the sublist narrows. So the cost grows with both the number of blocks back to the least-recently-active source (worst case the **entire store**) and the number of sources — all under `mset.mu` .

### The reverse block scan: rebuild each source's last sequence from stored headers

startingSequenceForSources walks back from LastSeq, LoadPrevMsgMulti loads (+decompresses) a block at a time until every source is found.



#### Why it can be slow on a limits-based stream

A single quiet/sparse source (S3) forces the scan past every newer block — worst case the entire stream. Each uncached block is read from disk, decrypted and decompressed. With many sources and a large store, leader election stalls under mset.mu while this runs.

## 2. The key insight

The store **already maintains a per-subject index** that can return the last sequence for a subject *without* walking messages:

- `fs.psim` — per-subject info, including `lblk` = the **last block index** that contains each subject ( `filestore.go:3843-3852` , `8980-8984` ).
- `mb.fss` — per-block subject state with `Last` per subject ( `filestore.go:9003-9015` ).

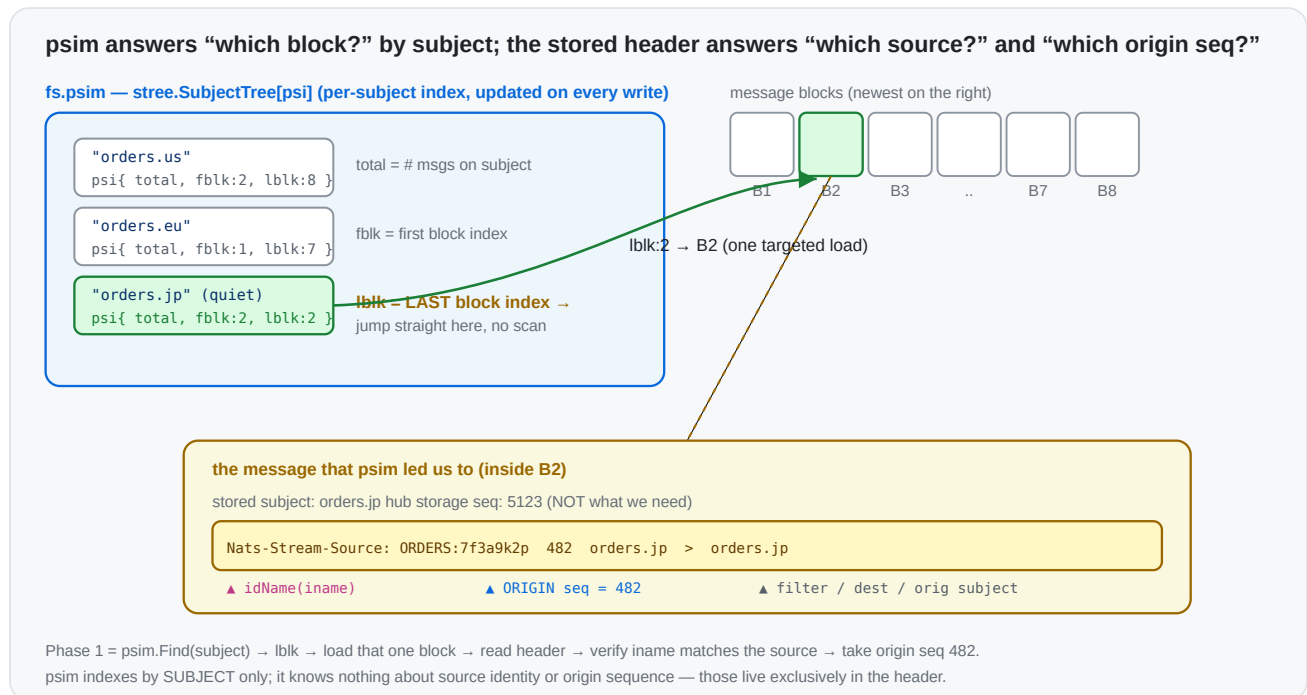
This is what powers `LoadLastMsg(subject) → loadLast ( filestore.go:8939 )`: for a literal subject it jumps straight to `info.lblk` and reads that subject's last message — **one targeted block load**, regardless of how far back it is. `MultiLastSeqs ( filestore.go:3806 )` does the same for a batch of filters in a single index pass.

The current scan ignores this index and instead does a generic backward message walk. The header it needs (the **origin** sequence) is right there in the message that `LoadLastMsg` already returns.

Why a plain "last seq for subject" isn't enough on its own: `si.sseq` must be the **origin** stream sequence taken from the `Nats-Stream-Source` header, not the hub's storage sequence. So we use the index to *find* the source's last stored message in  $O(1)$ , then read the origin seq from its header.

### 3. Background: the subject index ( `psim` + `fss` ) and the source header

The filestore keeps a **two-level subject index**. Phase 1 stands entirely on it, so it's worth being precise about what each level is.



#### 3.1 `psim` — the store-wide “which blocks hold this subject” map

`fs.psim` ( `filestore.go:195` ) is a `stree.SubjectTree[psi]` — a subject-token tree keyed by the full message subject — whose value is a tiny per-subject record ( `filestore.go:168` ):

```
type psi struct {
    total uint64 // how many messages exist for this subject across the whole s
    fblk  uint32 // index of the FIRST block that still contains this subject
    lblk  uint32 // index of the LAST block that still contains this subject
}
```

It is maintained incrementally on every write ( `filestore.go:4933-4943` ): on store, the subject's entry is created ( `total:1, fblk=lbld=current block` ) or its `total` is bumped and `lblk` advanced to the current block; on removal/expiry the counts and `fblk` / `lblk` are walked forward as blocks empty out. Because `psim` is a *subject tree*, it supports both exact `Find(subject)` and wildcard `Match(filter)` in time proportional to the subject token structure — **not** to the number of messages.

The single field that makes Phase 1 cheap is `lblk`: given a source's subject, `psim.Find` returns the **exact last block** that contains it. There is no need to scan newer blocks — most of which don't contain that subject at all. ( `MultiLastSeqs` , `filestore.go:3806` , uses the same `psim` to answer a *batch* of filters in one index pass.)

### 3.2 `fss` — the per-block "first/last seq of this subject in this block" map

`psim` gets us to a block; `mb.fss` ( `filestore.go:239` ) finishes the job *inside* that block. Each `msgBlock` carries its own `stree.SubjectTree[SimpleState]` mapping subject → ( `store.go:180` ):

```
type SimpleState struct {
    Msgs  uint64 // messages for this subject in THIS block
    First uint64 // first sequence for this subject in this block
    Last  uint64 // last  sequence for this subject in this block
    // (firstNeedsUpdate / lastNeedsUpdate: lazily recomputed when interior dele
}
```

So the two levels compose into an O(1)-ish lookup, which is exactly what `loadLast` ( `filestore.go:8980-9034` ) does:

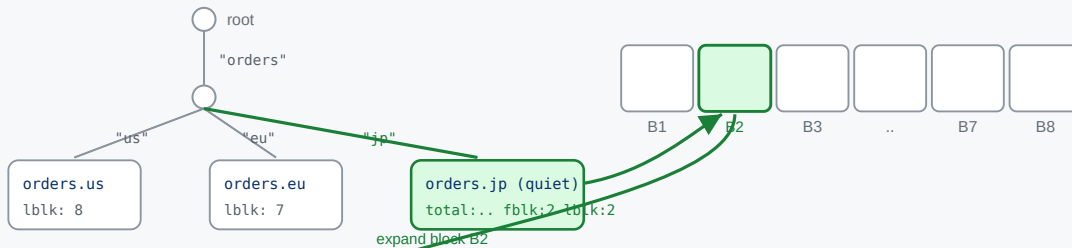
```
psim.Find(subject) → info.lblk → block → block.fss.Find(subject) → ss.Last
```

`fss` is what gives us the precise sequence within the block (and it correctly skips interior-deleted messages via `lastNeedsUpdate` / `recalculateForSubj` ). On a cold block, `ensurePerSubjectInfoLoaded` loads/derives `fss` for that **one** block — the cost we pay once per resolved source, versus the current scan paying it for every block back to the oldest source.

Two-level traversal: psim subject tree → the block → that block's fss subject tree → the message

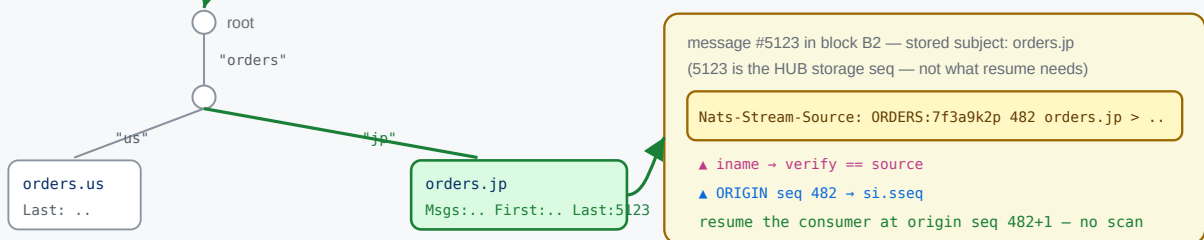
1 fs.psim — store-wide subject tree (subject → psi{total, fblk, lblk})

2 follow lblk → the last block holding that subject



3 B2.fss — per-block subject tree (subject → SimpleState{Msgs, First, Last})

4 load msg at Last (hub seq 5123) → read its header



### 3.3 Why each message carries a `Nats-Stream-Source` header

psim / fss index by **subject**. They know nothing about *which source* produced a stored message or *where in the origin stream* it came from — and those are precisely the two facts resume needs. That information is carried only in the per-message header `Nats-Stream-Source` (`JSStreamSource`, `stream.go:635`), written by `genSourceHeader` (`stream.go:4470`) as:

```
Nats-Stream-Source: <idName> <originSeq> <filter> <destination> <origSubject>
                    |           |
                    |           └ the ORIGIN stream's sequence – what we must
                    └ origin stream name (+ domain/consumer hash); with filter
                        reconstructs the source's full iname (parsed by stream)
```

This header is needed for three independent reasons:

1. **Source attribution.** A hub message is stored under its *destination* subject; the storage layer records nothing about its origin. When two sources can land on the same/overlapping subject, only the header's `iname` says which source a given message belongs to — hence Phase 1's `iname == si.iname` check.
2. **Origin-sequence tracking.** The sourcing consumer is (re)created on the **origin** with `DeliverByStartSequence`, which lives in the *origin's* sequence space. The hub's own storage sequence (e.g. 5123) is unrelated to the origin sequence (e.g. 482). Only the

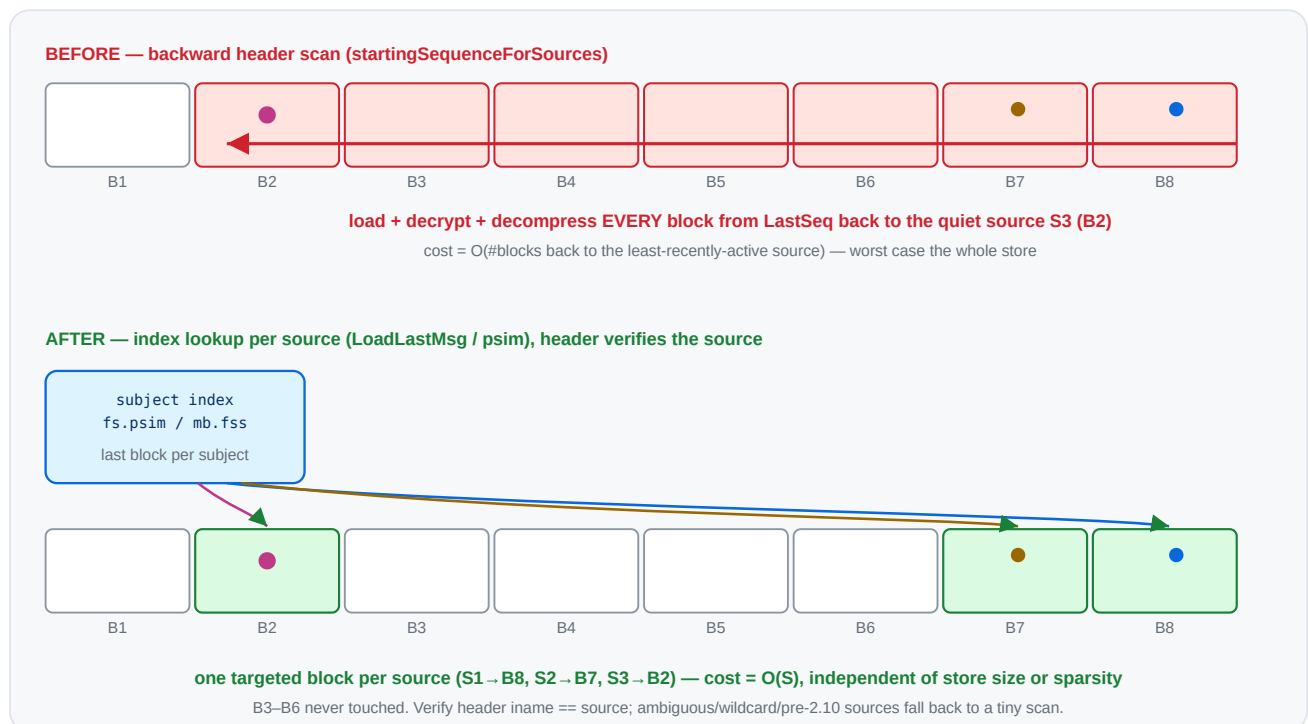
header preserves the origin sequence, so it is the *only* place a resume point can be recovered from without contacting the origin.

3. **Loop / daisy-chain handling & dedup.** When re-sourcing ( $A \rightarrow B \rightarrow C$ ), `processInboundSourceMsg` strips the inbound `JSSource` and stamps its own (`stream.go:4398`, `4405`), so each hop's provenance is well-defined and re-sourced duplicates can be recognised.

This is the crux of the whole proposal: **the index tells us *which block* in  $O(1)$ ; the header (read from that one message) tells us *which source* and *which origin seq*.** Today's scan reads the header off *every* message on the way back because it never consults `psim / fss` to jump directly to the right one.

## 4. Proposed algorithm

Two phases. Phase 1 resolves the common case with index lookups; Phase 2 keeps today's scan only for sources that genuinely can't be attributed by subject.



For each source, its **stored subject** is the `FilterSubject` for a plain source, or — for a source with a **single, concrete (non-templated)** subject transform — the transform **Destination**. Templated transform destinations contain `{{...}}` mapping tokens (the rendered subject varies per message), and partial wildcards in a destination are rejected at config time, so a destination is concrete unless it is empty, `>`, or contains `{{`.

Phase 1 – index lookup (per source,  $O(1)$  targeted load):  
for each source `si`:

```

    subj := si.FilterSubject // plain source
    if si has exactly one transform: subj := transform.Destination
    if subj == "" || subj == ">" || subj contains "{": defer to Phase 2; continue
    sm := store.LoadLastMsg(subj) // jumps via psim, ~1 block
    if sm has a JSStreamSource header:
        (_, iname, sseq) := streamAndSeq(header)
        if iname == si.iname: // self-verifying attribution
            si.sseq = sseq; mark resolved
    // anything not resolved (no header / header for a different source / wildcard
    // multi-or-templated transform / pre-2.10) → Phase 2

```

Phase 2 – fallback reverse scan (only for the unresolved subset):

run today's LoadPrevMsgMulti loop, but with the sublist built from ONLY the unresolved sources, so it narrows immediately and stops as soon as they are found

The header check makes Phase 1 **self-correcting**: if a subject is shared between sources, or overlaps a direct publish on the hub, or the last message has no source header, the lookup simply doesn't match and that source drops to Phase 2 — never producing a wrong answer.

## 5. Before → after (code)

**Before ( startingSequenceForSources , condensed — stream.go:4694 )**

```

mset.resetSourceInfo()
// build a sublist of ALL sources' subjects
refreshSublist() // every source
for last := state.LastSeq; ; { // walk the WHOLE store backwards
    sm, seq, err := mset.store.LoadPrevMsgMulti(sl, last, &smv) // loads/decompresses
    if err == ErrStoreEOF || err != nil { break }
    last = seq - 1
    ...
    _, iName, sseq := streamAndSeq(bytesToString(ss))
    update(iName, sseq) // narrows sublist as sources are found
    if len(seqs) == expected { return } // stop only when ALL sources are found
}

```

**Cost:** O(blocks back to the least-recently-active source) block loads → worst case the entire store.

## After

```
mset.resetSourceInfo()
var smv StoreMsg
unresolved := map[string]*sourceInfo{}

// Phase 1: targeted index lookup per source.
for _, ssi := range mset.cfg.Sources {
    si := mset.sources[ssi.iname]
    if si == nil { continue }
    var subj string // FilterSubject, or single c
    switch {
    case len(ssi.SubjectTransforms) == 0: subj = ssi.FilterSubject
    case len(ssi.SubjectTransforms) == 1: subj = ssi.SubjectTransforms[0].Destin
    }
    if subj == _EMPTY_ || subj == fwcs || strings.Contains(subj, "{{") {
        unresolved[ssi.iname] = si // can't attribute by a single
        continue
    }
    sm, err := mset.store.LoadLastMsg(subj, &smv) // 0(1) via psim/fss, returns
    if err != nil || sm == nil || len(sm.hdr) == 0 {
        unresolved[ssi.iname] = si
        continue
    }
    if ss := sliceHeader(JSStreamSource, sm.hdr); len(ss) > 0 {
        if _, iName, sseq := streamAndSeq(bytesToString(ss)); iName == ssi.iname {
            si.sseq, si.dseq = sseq, 0 // verified – resolved without
            continue
        }
    }
    unresolved[ssi.iname] = si
}

// Phase 2: only the leftovers pay the (now tiny) reverse scan.
if len(unresolved) > 0 {
    mset.reverseScanForSources(unresolved) // = today's loop, scoped to
}
```

**Cost:** Phase 1 =  $O(S)$  targeted lookups (recent sources share/cache the last block; a quiet source is **one** targeted load, not a walk). Phase 2 only runs for ambiguous sources, with a smaller sublist.



Multi-destination transform sources: call `LoadLastMsg` for each destination and keep the highest origin `sseq`, or defer them to Phase 2. Either is fine; deferring is simplest to start.

## 6. Before → after (complexity)

|                                      | Before                                           | After                                                                                            |
|--------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Index used                           | none (generic message walk)                      | <code>psim</code> / <code>fss</code> per-subject last-block index                                |
| Blocks loaded                        | back to least-active source ( $\leq$ all blocks) | ~1 per source, only the blocks actually holding a source's last msg                              |
| Sensitivity to a <b>quiet</b> source | drags the scan back across the whole store       | none — jumps straight to its block                                                               |
| Sensitivity to <b>store size</b>     | linear in blocks scanned                         | none                                                                                             |
| Worst case still possible?           | normal case                                      | only if every source is <code>&gt; /wildcard/shared-subject/pre-2.10</code> (→ Phase 2 == today) |
| Held lock                            | <code>mset.mu</code> for the whole walk          | <code>mset.mu</code> for O(S) lookups; Phase 2 only for leftovers                                |

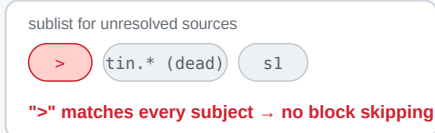
The edge → hub topology (each edge mapped to a distinct subject/domain) lands entirely in Phase 1 → the backward scan disappears for that case.

## 7. Phase 2: keeping the fallback scan narrow

Phase 1 removes most sources from the picture, but whatever remains still runs the reverse `LoadPrevMsgMulti` scan. That scan's whole speed advantage depends on the **sublist being narrow** — `prevMatchingMulti` uses each block's `fss` to skip blocks that contain none of the sublist's subjects. A single `>` (full wildcard) in the sublist defeats that: it matches every subject, so no block can be skipped and the scan visits every block back to the match.

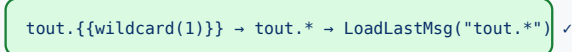
## Phase 2 sublist narrowing: one transform source no longer poisons the whole scan

**BEFORE** — transform/catch-all source injects ">" (empty FilterSubject)



**AFTER** — phase 1 resolves wildcard transforms; phase 2 sublist uses destinations

step 1: phase 1 collapses wildcard(/)\$N tokens and resolves via the index



step 2: only genuinely-unresolved sources remain; sublist stays narrow



Remaining ">" cases (by design): a genuine catch-all source (empty filter, no transform) or an exotic transform (partition/split).

## What used to inject >

1. **Every transform source.** A transform source has an empty `FilterSubject` (mutually exclusive with `SubjectTransforms`, `stream.go:1976`), so the old `refreshSublist` did `s1.Insert(fwcs)`. One transform source in the unresolved set widened the whole sublist to `>` → full backward scan.
2. **Templated transforms were deferred to Phase 2 at all.** Their *rendered* destination is a narrow wildcard (e.g. `tout.{{wildcard(1)}}` always lands on `tout.*`), so they don't need a full scan.
3. **Dead entries.** Phase 2 also inserted `si.sfs` — the transform *source* filters (e.g. `tin.*`) — which never match stored subjects (messages are stored under the *destination*).

## The two changes

- **Resolve wildcard transforms in Phase 1.** `transformUntokenize(dest)` collapses `wildcard()` / `$N` mapping tokens to a subject wildcard (`tout.{{wildcard(1)}}` → `tout.*`); `LoadLastMsg` accepts that wildcard and the header- `iname` check still guards attribution. So the common templated transforms resolve in Phase 1 and never reach the scan.
- **Build the Phase 2 sublist from destinations.** For any transform source that *does* reach Phase 2 (e.g. its wildcard space is shared with another source, so Phase 1 saw a header for the other one), insert the destination's wildcard form instead of `>`, and drop the dead `si.sfs` inserts.

## What still forces > (by design)

Only a source that *genuinely* needs a broad match:

- a **catch-all source** — empty `FilterSubject`, no transform — sourcing every subject of its origin; and
- an **exotic transform** ( `partition / split / slice /...` ) whose rendered destination isn't a subject wildcard, so `transformUntokenize` can't reduce it (it leaves the `{{...}}` token in place, which we detect and fall back to `>` ).

Both are uncommon for the edge → hub fan-in. Distinct subjects, wildcard filters, and `wildcard() / $N` transforms all either resolve in Phase 1 or keep the Phase 2 sublist narrow enough to skip blocks.

Correctness note: this is purely about *narrowing*, never about *missing* a source. If a narrowed subject is wrong (shared space), the header check fails and the source stays in the unresolved set; an exotic/empty case falls back to `>` . So Phase 2 still finds every source it did before — it just loads far fewer blocks getting there.

## 8. Correctness & edge cases

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- **Same semantics.** Today's scan records, per source, the **most recent** stored message's origin seq (first hit scanning backward). `LoadLastMsg` returns exactly that message. Deleted/interior messages are handled by `loadLast` (it skips `dmap` entries and walks to the previous block if needed), matching the scan's behaviour.
- **Shared subjects / direct publishes.** Resolved safely by the header-iname verification → fall to Phase 2.
- **Wildcard / empty ( `>` ) filters.** Can't be attributed to one source by subject → Phase 2 (same as today for them).
- **Pre-2.10 headers** (no `iname`, only stream name): verification won't match → Phase 2, which keeps the existing stream-name matching path.
- **memstore.** `MemStore` has the simpler linear `LoadPrevMsgMulti`; `LoadLastMsg` there is also index-light, but memstore streams are small, so Phase 2 cost is negligible. (We can add a memstore `loadLast` fast path later if needed.)
- **setStartingSequenceForSources** (the `STREAM.UPDATE` path, `stream.go:4566` ) gets the same Phase 1 treatment for the subset of sources it processes.

## 9. Risks

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- **Index freshness.** `psim / fss` may lazily need a recalculation ( `lastNeedsUpdate`, `recalculateForSubj` ); `loadLast / MultiLastSeqs` already handle that, so we inherit

correct behaviour.

- **Behavioural parity.** Phase 1 must produce identical `si.sseq` to the old scan for non-ambiguous sources; this is the core test (below). Keeping Phase 2 byte-for-byte as today bounds the blast radius.
- **Encryption/compression.** Phase 1 still loads the one block holding a source's last message (decrypt/decompress), but only that block — no change in correctness, large reduction in volume.

## 10. Testing & validation

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### Prototype status — implemented

A working prototype of the two-phase resolver is implemented in `startingSequenceForSources` ( `server/stream.go` ), with:

- `TestJetStreamStartingSequenceForSourcesIndexFastPath` ( `server/jetstream_sourcing_resume_test.go` ) — three distinct-subject sources (index fast path) plus one **subject-transform** source (phase 2 fallback), each with a different origin depth and buried under 20k direct publishes. Asserts every recovered `si.sseq` equals the expected last origin sequence. **Passes.**
- `TestJetStreamStartingSequenceForSourcesAmbiguity` — a catch-all (empty filter) source and **two sources transformed onto the same destination subject** (shared stored subject), mixed with a distinct-subject source. Asserts the phase 2 fallback disambiguates all of them. **Passes.**
- `TestJetStreamSetStartingSequenceForSourcesIndex` — the same fast path applied to the `STREAM.UPDATE` twin `setStartingSequenceForSources` (distinct subject + catch-all + a **concrete-destination transform** source), clearing and recovering `sseq`. **Passes.**
- **Transform fast path:** a source with a single subject transform is resolved in Phase 1 by collapsing `wildcard()` / `$N` mapping tokens to a wildcard form ( `transformUntokenize` , e.g. `tout.{{wildcard(1)}}` → `tout.*` ) and doing `LoadLastMsg(wildcardForm)` , verified by the header. Only multi-transform and **exotic** transforms ( `partition` / `split` / ... that don't reduce to a subject wildcard) still defer to Phase 2.
- **Phase 2 no longer degrades to `>` for transform sources.** Previously a transform source (empty `FilterSubject` ) inserted a `>` catch-all into the sublist, defeating per-block skipping for the whole scan. The Phase 2 sublist is now built from the transform **destinations** (wildcard form), with the dead `si.sfs` source-filter inserts removed; only a genuine catch-all source or an exotic transform still forces `>` .
- The change was also applied to `setStartingSequenceForSources` itself ( `stream.go:4566` ), so both the leader-election and the config-update resume paths use the index fast path and the narrowed sublist.

- Tests (all pass, incl. `-race`): `...IndexFastPath`, `...Ambiguity` (catch-all + shared concrete destination), `...TemplatedTransform` (wildcard transform), `...SharedWildcardTransform` (two templated transforms sharing a `gout.*` space), and `TestJetStreamSetStartingSequenceForSourcesIndex` (twin).
- `BenchmarkJetStreamScanForSources` (existing, single source) and a new `BenchmarkJetStreamScanForSourcesMulti` (16 sources spread across the store).
- Existing sourcing suite ( `SourceBasics`, `SourceRemovalAndReAdd`, `WorkQueueSourceRestart`, `SourceWorkingQueueWithLimit`, `StreamSourceWithoutDuplicateWindow`, `MirrorAndSourcesFilteredConsumers` ) still passes.

## Measured (filestore, single-server)

| Benchmark                                                 | Before (reverse scan) | After (phase 1) | Speedup |
|-----------------------------------------------------------|-----------------------|-----------------|---------|
| <code>ScanForSources</code> — 1 source, ~100k buried msgs | ~8.5 µs/op            | ~5.5 µs/op      | ~1.5×   |
| <code>ScanForSourcesMulti</code> — 16 sources, spread     | ~110.8 µs/op          | ~14.0 µs/op     | ~7.9×   |

The single-source gap is modest because `LoadPrevMsgMulti` already skips non-matching messages per block; the win scales with the number of sources (each avoids a re-walk via a direct `psim` jump), which is exactly the edge → hub fan-in case.

## What still forces a full `>` scan in Phase 2 (by design)

After the above, Phase 2 only widens to `>` (no block skipping) when a source *genuinely* needs it:

- a **catch-all source** — empty `FilterSubject`, no transform — that sources every subject of its origin (it really can land anywhere, so a broad match is correct); and
- an **exotic transform** ( `partition` / `split` / `slice` /...) whose rendered destination is not a subject wildcard, so `transformUntokenize` can't reduce it to a matchable pattern.

Both are uncommon for the edge → hub fan-in. Everything else — distinct subjects, wildcard filters, `wildcard()` / `$N` transforms — either resolves in Phase 1 or narrows the Phase 2 sublist to a concrete or wildcard subject.

## Still to do

- **Pre-2.10 / direct-publish-overlap coverage:** add explicit cases (handled today via the header check and the Phase 2 fallback; tests would lock the behaviour in). Hard to

construct through the JS client because streams can't declare overlapping subjects; would need low-level store seeding.

- A `partition / split` transform benchmark to confirm the (rare) `>` path is acceptable.

## Resolved finding (was suspected pre-existing bug)

An earlier note suspected `setStartingSequenceForSources` mishandled transform sources because its Phase 2 sublist used `si.sfs` (transform **source** filters) rather than the **destination**. On inspection this was **not** a correctness bug: transform sources have an empty `FilterSubject`, so the old sublist inserted a `>` catch-all that matched the destination anyway (verified by `TestJetStreamStartingSequenceForSourcesTemplatedTransform`). It was purely an **efficiency** problem — the `>` defeated block skipping — now fixed by building the Phase 2 sublist from the destination wildcard form and dropping the dead `si.sfs` inserts.

## 11. Relationship to the durable-consumer proposal

This change is **orthogonal and complementary** to the durable-consumer/ `si.sseq` -persistence ideas in `sourcing-durable-resume.md`:

- Durable consumers / persisted `si.sseq` aim to **skip** resume work entirely (when state survives).
- This proposal makes the **fallback resume itself cheap** — which still runs on cold start, on snapshot restore, for limits/clustered streams, and any time persisted state is missing.

Recommended order: land this index-based resume first (self-contained, no protocol/state change, helps every retention type today), then layer the persistence/durable improvements on top.

## Appendix — key references

| Symbol                                     | File:line                   | Role                                    |
|--------------------------------------------|-----------------------------|-----------------------------------------|
| <code>startingSequenceForSources</code>    | <code>stream.go:4694</code> | the scan to replace (Phase 1 + Phase 2) |
| unconditional call                         | <code>stream.go:4821</code> | runs on every leader election / restart |
| <code>setStartingSequenceForSources</code> | <code>stream.go:4566</code> | update-path twin, same treatment        |

| Symbol                                                   | File:line                                                      | Role                                                                             |
|----------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------|
| sublist build (stored subjects)                          | <code>stream.go:4742-4757</code>                               | defines a source's stored subject(s)                                             |
| <code>LoadLastMsg</code> / <code>loadLast</code>         | <code>filestore.go:9048</code> / <code>8939</code>             | index-based last-msg-for-subject (returns header)                                |
| <code>MultiLastSeqs</code>                               | <code>filestore.go:3806</code>                                 | batched last-seq-per-filter via index                                            |
| <code>psi</code> / <code>fs.psim</code>                  | <code>filestore.go:168</code> / <code>195</code>               | per-subject <code>{total, fblk, lblk}</code> ; store-wide subject → blocks index |
| <code>mb.fss</code> / <code>SimpleState</code>           | <code>filestore.go:239</code> / <code>store.go:180</code>      | per-block subject → <code>{Msgs, First, Last}</code> index                       |
| <code>psim</code> maintenance on write                   | <code>filestore.go:4933-4943</code>                            | how <code>total</code> / <code>lblk</code> are kept current                      |
| <code>LoadPrevMsgMulti</code>                            | <code>filestore.go:9403</code> / <code>memstore.go:1991</code> | the backward walk used by Phase 2                                                |
| <code>Nats-Stream-Source</code> header                   | <code>stream.go:635</code>                                     | constant; the per-message source provenance                                      |
| <code>genSourceHeader</code> / <code>streamAndSeq</code> | <code>stream.go:4470</code> / <code>4545</code>                | writes / parses origin stream, iname, origin seq                                 |